Task 3: Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

```
 GNU nano 7.2                                    process_state.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Before fork():\n");
    printf("PID  = %d\n", getpid());
    printf("PPID = %d\n", getppid());
    printf("State = Parent process is running\n\n");

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    else if (pid == 0) {
        printf("Child Process:\n");
        printf("PID  = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("State = Child process is running\n\n");

        sleep(2);

        printf("Child Process After Sleep:\n");
        printf("PID  = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("State = Child process completed\n");

        exit(0);
    }

    else {
        printf("Parent Process:\n");
        printf("PID  = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("State = Parent waiting for child\n\n");

        wait(NULL);

        printf("Parent Process After wait():\n");
        printf("PID  = %d\n", getpid());
        printf("PPID = %d\n", getppid());
        printf("State = Child completed, parent continues\n");
    }

    return 0;
}
```

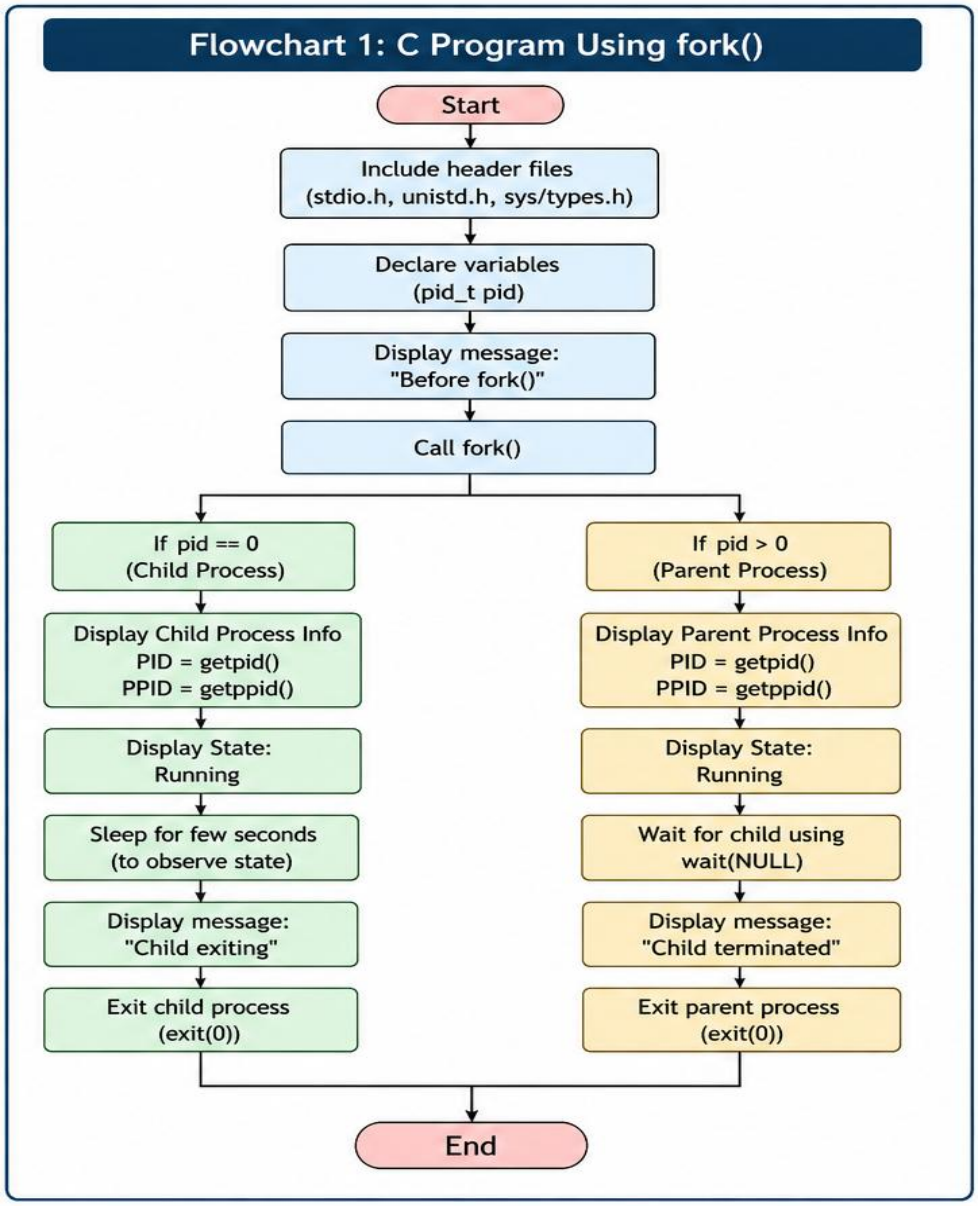
```
varshini@DESKTOP-BLT5L3R:~$ cd ~
varshini@DESKTOP-BLT5L3R:~$ nano process_state.c
varshini@DESKTOP-BLT5L3R:~$ gcc process_state.c -o process_state
varshini@DESKTOP-BLT5L3R:~$ ./process_state
Before fork():
PID  = 820
PPID = 332
State = Parent process is running

Parent Process:
PID  = 820
PPID = 332
State = Parent waiting for child

Child Process:
PID  = 821
PPID = 820
State = Child process is running

Child Process After Sleep:
PID  = 821
PPID = 820
State = Child process completed
Parent Process After wait():
PID  = 820
PPID = 332
State = Child completed, parent continues
varshini@DESKTOP-BLT5L3R:~$
```

Flow Chart



Observation

1. fork() successfully creates a **parent and child process** with different PIDs.
2. The child process has the **parent's PID as its PPID**.
3. Process states change as the parent waits and the child executes.
4. wait() synchronizes the parent with the completion of the child process.

Task 4: Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

```
 GNU nano 7.2                          state.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("Process started\n");
    printf("PID = %d\n", getpid());
    fflush(stdout);

    printf("Running...\n");
    sleep(10);

    printf("Waiting...\n");
    sleep(20);

    printf("Running again...\n");
    sleep(10);

    printf("Process terminating...\n");

    return 0;
}
```

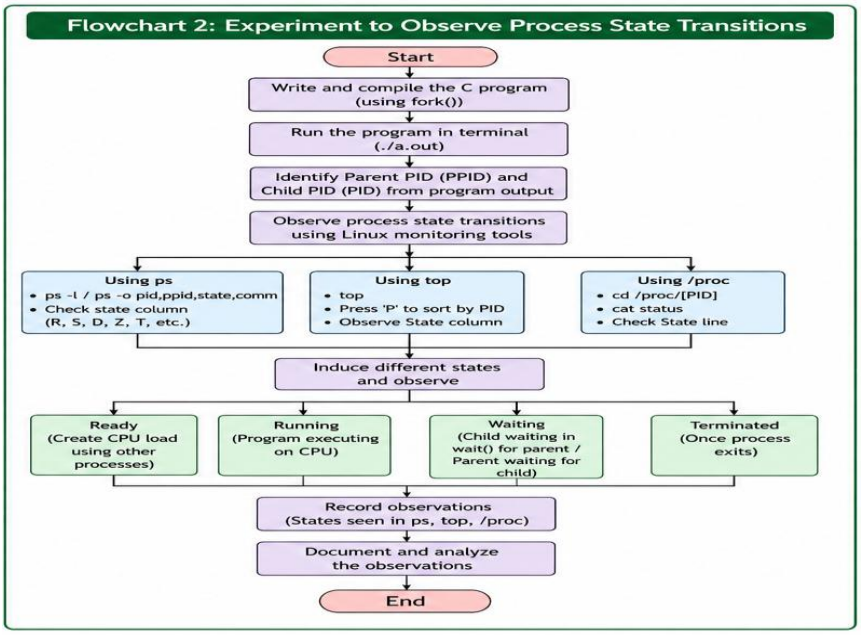
```
varshini@DESKTOP-BLT5L3R:~$ nano state.c
varshini@DESKTOP-BLT5L3R:~$ gcc state.c -o state
varshini@DESKTOP-BLT5L3R:~$ ./state &
[1] 876
varshini@DESKTOP-BLT5L3R:~$ Process started
PID = 876
Running...
Waiting...
Running again...
Process terminating...
```

```
varshini@DESKTOP-BLT5L3R:~$ ps -o pid,ppid,state,stat,cmd -p 3456
    PID    PPID S STAT CMD
varshini@DESKTOP-BLT5L3R:~$ top -p 3456
top - 06:48:30 up 52 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:   0 total,   0 running,   0 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7784.6 total,   7139.8 free,    532.2 used,    263.1 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7252.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
```

```
varshini@DESKTOP-BLT5L3R:~$ pstree -p
systemd(1)─┬─agetty(227)
           ├─cron(186)
           ├─dbus-daemon(187)
           ├─init-systemd(Ub(2)─┬─SessionLeader(916)───Relay(922)(918)───ba+
           │                    ├─init(8)───{init}(9)
           │                    ├─login(333)───bash(412)
           │                    └─{init-systemd(Ub}(10)
           ├─rsyslogd(217)─┬─{rsyslogd}(235)
           │               ├─{rsyslogd}(236)
           │               └─{rsyslogd}(237)
           ├─systemd(383)───(sd-pam)(384)
           ├─systemd-journal(63)
           ├─systemd-logind(200)
           ├─systemd-resolve(173)
           ├─systemd-timesyn(177)───{systemd-timesyn}(181)
           ├─systemd-udevd(109)
           ├─unattended-upgr(244)───{unattended-upgr}(287)
           └─wsl-pro-service(203)─┬─{wsl-pro-service}(247)
                                  ├─{wsl-pro-service}(248)
                                  ├─{wsl-pro-service}(249)
                                  ├─{wsl-pro-service}(250)
                                  ├─{wsl-pro-service}(251)
                                  ├─{wsl-pro-service}(253)
                                  └─{wsl-pro-service}(254)
varshini@DESKTOP-BLT5L3R:~$ ps -p 3456
    PID TTY          TIME CMD
varshini@DESKTOP-BLT5L3R:~$
```

Flow Chart:



Observation

1. The process state can be monitored using **ps, top, and /proc**.
2. The process changes between **running/runnable and waiting/sleeping states** during execution.
3. After completion, the process enters the **terminated state** and disappears from the process table and /proc.
4. Linux represents process states using codes such as **R (running/runnable) and S (sleeping/waiting)**.